Comprehensive Spatio-Temporal Analysis and Predictive Modeling of Precipitation Events Across the United States (2016–2022)

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***Abstract*—This paper presents an exhaustive spatio-temporal analysis of precipitation-related weather events across 2,071 meteorological stations spanning 49 U.S. states over a seven-year period (January 2016–December 2022). We employ a comprehensive analytical framework integrating global and local spatial autocorrelation measures (Moran's I, LISA, Getis-Ord G*), geostatistical techniques (variography, spatial correlograms), kernel density estimation for pattern visualization, and predictive modeling using Ordinary Least Squares (OLS), Geographically Weighted Regression (GWR), and Random Forest algorithms with rigorous spatial holdout validation. Our analysis reveals weak but statistically significant global spatial autocorrelation (I=0.0177, p¡0.0001), substantial spatial heterogeneity at kilometer scales, and distinct temporal patterns within identified spatial clusters. The Random Forest model significantly outperforms linear approaches in spatial prediction tasks, with event severity emerging as the dominant predictor (48% feature importance). We provide a fully reproducible pipeline (`stda.py`), comprehensive visualizations, and detailed methodological justifications. This work contributes to the growing body of literature on spatial environmental data science by demonstrating the complementary nature of exploratory spatial statistics and predictive machine learning in understanding complex meteorological phenomena.**

***Index Terms*—spatial autocorrelation, LISA, Getis-Ord G*, variogram, correlogram, kernel density estimation, geographically weighted regression, random forest, spatial validation, precipitation patterns, spatio-temporal analysis**

I. Introduction

A. Background and Motivation

Spatially explicit environmental datasets have become increasingly available due to advances in sensor networks, satellite observations, and data logging infrastructure. Weather-station-derived precipitation events represent a canonical example of spatial point data with associated attributes (location, timestamp, intensity, event type). Understanding the spatial structure and temporal dynamics of such phenomena is critical for climate monitoring, disaster preparedness, water resource management, and ecological modeling [4].

The United States benefits from extensive meteorological monitoring infrastructure, with thousands of automated weather stations recording hourly observations. This dense spatial coverage, combined with multi-year temporal extent, enables sophisticated spatial statistical analyses that can reveal clustering patterns, identify hotspots, characterize spatial heterogeneity, and support predictive modeling for unobserved locations.

B. Research Questions

This study addresses the following research questions:

1) **RQ1:** Does precipitation exhibit significant spatial autocorrelation at the national scale, and what is the nature of this spatial dependence?
2) **RQ2:** Where are the significant local clusters (hotspots and coldspots) of precipitation intensity located?
3) **RQ3:** What is the characteristic spatial range of precipitation correlation, and how does it decay with distance?
4) **RQ4:** How do identified spatial clusters evolve temporally across the 2016–2022 period?
5) **RQ5:** Can spatial and temporal features effectively predict precipitation at unobserved locations, and which modeling approaches perform best?
6) **RQ6:** What are the most important predictive features, and how do they vary spatially?

C. Contributions

Our primary contributions are:

1) A reproducible end-to-end analytical pipeline (`stda.py`) implementing preprocessing, spatial weights construction, and comprehensive spatial statistical analyses
2) Rigorous application of complementary spatial methods (global/local autocorrelation, geostatistics, density estimation) with detailed interpretation
3) Spatial prediction experiments using OLS, GWR, and Random Forest with proper spatial holdout validation to avoid optimistic bias
4) Temporal analysis of spatial cluster dynamics, revealing differential trends across LISA quadrants
5) Extensive visualization suite (16 figures) documenting all analysis phases
6) Critical assessment of method suitability, limitations, and recommendations for future work

D. Paper Organization

The remainder of this paper is structured as follows: Section II describes the dataset and preprocessing; Section III details the methodological framework; Section IV presents experimental setup and validation strategy; Section V comprehensively reports results with extensive visual documentation; Section VI discusses findings, method performance, and limitations; Section VII concludes and identifies future research directions.

II. Dataset and Preprocessing

A. Data Description

The analyzed dataset comprises weather event records from 2,071 airport-based meteorological stations distributed across 49 U.S. states (all states except Hawaii). The raw dataset contains approximately 8.6 million event records spanning January 1, 2016 through December 31, 2022 (7 years). Each record includes:

- Spatial coordinates: latitude and longitude (WGS84)
- Temporal information: start time and end time (UTC)
- Event attributes: type (rain, snow, fog, etc.), severity classification
- Numeric measurements: precipitation amount (inches), visibility, temperature
- Station metadata: airport code, elevation, location name

The dataset exhibits substantial spatial coverage with reasonable station density in populated regions, though coverage is sparser in remote areas (mountain ranges, desert regions). Temporal coverage is consistent across the study period with minimal gaps.

B. Data Quality Assessment

Initial exploratory data analysis revealed several data quality issues requiring attention:

- Missing coordinates: 0.3% of records lacked valid lat/lon values
- Duplicate records: 1.2% exact duplicates based on all fields
- Temporal duplicates: Multiple records for the same event ID with different timestamps (retained most recent)
- Outliers: Some extreme precipitation values (¿20 inches) verified as legitimate extreme events
- Missing values: Approximately 5% missing values in secondary attributes (not affecting primary analysis)

C. Preprocessing Pipeline

The preprocessing workflow implemented in `stda.py` consists of:

Step 1: Temporal Parsing

- Parse `StartTime(UTC)` and `EndTime(UTC)` to pandas datetime objects
- Extract temporal features: year, month, day-of-week (0=Monday), hour, weekend indicator
- Calculate event duration in minutes

Step 2: Coordinate Validation and Cleaning

- Remove records with missing latitude or longitude
- Validate coordinate ranges: lat $\in$ [24°N, 49°N], lon $\in$ [-125°W, -66°W]
- Remove obvious geocoding errors (e.g., swapped lat/lon)

Step 3: Deduplication

- Sort by event ID and timestamp
- Keep most recent record for each unique event ID
- Remove exact duplicate rows across all fields

Step 4: Spatial Data Structure Creation

- Create GeoDataFrame using GeoPandas with point geometries
- Initial CRS: EPSG:4326 (WGS84 geographic coordinates)
- Reproject to EPSG:3857 (Web Mercator) for metric distance calculations
- Extract projected X and Y coordinates for analysis

Step 5: Target Variable Selection and Transformation

- Identify primary analysis variable: `Precipitation(in)`
- Apply log transformation: `Precipitation(in)_log` = log(1 + Precipitation)
- Log transformation addresses right-skewness and stabilizes variance
- Handle zero and negative values appropriately

Step 6: Feature Engineering

- Encode categorical variables: event type, severity level
- Create spatial lag features (k-nearest neighbor average of target)
- Calculate distance to geographic centroid
- Derive interaction terms for exploratory modeling

D. Final Dataset Statistics

After preprocessing, the analysis dataset contains:

- 8,437,291 valid event records
- 2,071 unique spatial locations (stations)
- Temporal range: 2,556 days (7 full years)
- Target variable range: Precipitation(in)_log $\in$ [0, 1.65]
- Mean precipitation (log): 0.731, Std: 0.089

III. Methodology

This section provides mathematical foundations and implementation details for all analytical methods employed.

A. Spatial Weights Construction

Spatial weights matrices $\mathbf{W}$ formalize the neighborhood structure used in spatial statistics. We employed K-Nearest Neighbors (KNN) weights:

KNN Weights: For each location i, identify the k nearest neighbors based on Euclidean distance in projected coordinates. The weight is:

$$w_{ij} = \begin{cases} 1/k & \text{if } j \in \text{KNN}(i) \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Row standardization ensures $\sum_j w_{ij} = 1$, making spatial lag interpretable as a local average.

Implementation: We used $k = 8$ neighbors as a balance between local context and computational efficiency. For computational reasons, spatial statistics were computed on subsamples (50,000 records for Moran's I, 5,000 for LISA/G*).

B. Global Spatial Autocorrelation: Moran's I

Moran's I measures the degree to which similar values cluster in space [1]:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \tag{2}$$

where:

- n: number of spatial units
- x_i: attribute value at location i
- $\bar{x}$: mean of attribute values
- w_{ij}: spatial weight between locations i and j

Interpretation:

- $I > 0$: positive autocorrelation (clustering of similar values)
- $I \approx 0$: random spatial pattern
- $I < 0$: negative autocorrelation (checkerboard pattern)

Statistical Significance: Assessed using permutation tests (999 random permutations) to generate null distribution under spatial randomness hypothesis.

C. Spatial Correlogram

A spatial correlogram plots Moran's I against distance, revealing the decay of spatial autocorrelation:

$$I(h) = \text{Moran's I for distance lag } h \tag{3}$$

Multiple distance bins (lags) are defined, and Moran's I is computed for each using pairs within that distance range. This reveals:

- Spatial range: distance beyond which $I(h) \approx 0$
- Shape of decay: exponential, linear, or oscillating patterns
- Anisotropy: directional dependence (if computed for directional lags)

D. Local Indicators of Spatial Association (LISA)

LISA decomposes global Moran's I into location-specific components [2]:

$$I_i = \frac{(x_i - \bar{x})}{m_2} \sum_j w_{ij}(x_j - \bar{x}) \tag{4}$$

where $m_2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$ is the variance.

LISA Quadrants: Significant I_i values are classified into:

- **High-High (HH):** High value surrounded by high values (hotspot)
- **Low-Low (LL):** Low value surrounded by low values (coldspot)
- **High-Low (HL):** High value surrounded by low values (spatial outlier)
- **Low-High (LH):** Low value surrounded by high values (spatial outlier)
- **Not Significant (ns):** $p > 0.05$ after multiple testing correction

Significance assessed via conditional permutation: hold x_i fixed, permute neighbors.

E. Getis-Ord G* Statistic

The G_i^* statistic identifies local clusters without distinguishing between high-high and low-low patterns [3]:

$$G_i^* = \frac{\sum_j w_{ij} x_j - \bar{x} \sum_j w_{ij}}{s\sqrt{\frac{n \sum_j w_{ij}^2 - (\sum_j w_{ij})^2}{n-1}}} \tag{5}$$

where s is the standard deviation of x.

Standardized Z-scores:

- $Z(G_i^*) > 0$: local cluster of high values (hot spot)
- $Z(G_i^*) < 0$: local cluster of low values (cold spot)
- $|Z(G_i^*)| > 1.96$: significant at $\alpha = 0.05$ (two-tailed)

F. Empirical Variogram

The variogram quantifies spatial variance as a function of distance [4]:

$$\hat{\gamma}(h) = \frac{1}{2|N(h)|} \sum_{(i,j) \in N(h)} (x_i - x_j)^2 \tag{6}$$

where $N(h)$ is the set of location pairs separated by lag distance h (within tolerance).

Variogram Components:

- **Nugget** (c_0): discontinuity at origin (measurement error or micro-scale variation)
- **Sill** ($c_0 + c$): plateau value (total variance)
- **Range** (a): distance where variogram reaches sill (spatial correlation length)

Model Fitting: We fit spherical models to empirical variograms:

$$\gamma(h) = \begin{cases} c_0 + c\left[\frac{3h}{2a} - \frac{h^3}{2a^3}\right] & h \le a \\ c_0 + c & h > a \end{cases} \tag{7}$$

G. Kernel Density Estimation (KDE)

KDE estimates the spatial intensity of point patterns:

$$\hat{f}(\mathbf{s}) = \frac{1}{n} \sum_{i=1}^{n} K_h(\mathbf{s} - \mathbf{s}_i) \tag{8}$$

where K_h is a kernel function with bandwidth h (here: Gaussian with $h = 2000$ meters).

Value-Weighted KDE: Incorporates attribute values:

$$\hat{f}_w(\mathbf{s}) = \frac{\sum_{i=1}^{n} w_i K_h(\mathbf{s} - \mathbf{s}_i)}{\sum_{i=1}^{n} w_i} \tag{9}$$

where w_i is the precipitation value at location i. This highlights regions with intense events, not just high event counts.

H. Regression Models

1) Ordinary Least Squares (OLS): Baseline linear model:

$$y = \beta_0 + \sum_{j=1}^{p} \beta_j x_j + \varepsilon \tag{10}$$

Estimated via:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{11}$$

2) Geographically Weighted Regression (GWR): GWR allows coefficients to vary spatially [5]:

$$y_i = \beta_0(u_i, v_i) + \sum_{j=1}^{p} \beta_j(u_i, v_i)x_{ij} + \varepsilon_i \tag{12}$$

Estimated via weighted least squares at each location i:

$$\hat{\boldsymbol{\beta}}(u_i, v_i) = (\mathbf{X}^T\mathbf{W}(u_i, v_i)\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}(u_i, v_i)\mathbf{y} \tag{13}$$

where $\mathbf{W}(u_i, v_i)$ is a diagonal weight matrix with Gaussian distance decay:

$$w_{ij} = \exp\left(-\frac{d_{ij}^2}{b^2}\right) \tag{14}$$

Bandwidth b selected via cross-validation (AICc minimization).

Advantages: Captures spatial non-stationarity in relationships.

Challenges: Computationally intensive; requires careful bandwidth selection; can overfit with small bandwidths.

3) Random Forest: Ensemble method using decision trees [7]:

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}) \tag{15}$$

where T_b is the b-th tree trained on bootstrap sample.

Hyperparameters:

- Number of trees: 100
- Max depth: 20
- Min samples per leaf: 5
- Features per split: $\sqrt{p}$

Feature Importance: Mean decrease in impurity across trees.

IV. Experimental Setup and Validation

A. Computational Considerations

Given the large dataset (8.4M records), we employed strategic subsampling:

- **Global Moran's I:** 50,000 randomly sampled records
- **LISA/G*:** 5,000 randomly sampled records (computationally intensive)
- **Variogram:** 10,000 records with spatial stratification
- **GWR bandwidth selection:** 5,000 records
- **GWR final model:** 20,000 records
- **Machine learning:** Full dataset with spatial cv

All random sampling used fixed seeds (42) for reproducibility.

B. Spatial Holdout Validation

Traditional random train-test splits create optimistic bias in spatial prediction due to spatial autocorrelation (nearby training points leak information to test points).

Solution: Spatial block cross-validation

1) Cluster locations into $K = 5$ spatial blocks using K-Means on coordinates
2) Iteratively hold out one block as test set
3) Train on remaining $K - 1$ blocks
4) Average performance across folds

This simulates prediction for geographically distinct regions.

C. Evaluation Metrics

Regression Metrics:

- $R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$: proportion of variance explained
- RMSE $= \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$: root mean squared error
- MAE $= \frac{1}{n}\sum|y_i - \hat{y}_i|$: mean absolute error

Spatial Autocorrelation of Residuals: Moran's I of residuals tests for remaining spatial structure (should be ≈ 0 for good models).

V. Results

A. Temporal Patterns

1) Overall Temporal Trend: Figure 1 presents the year-by-year trend of mean log-transformed precipitation. Key observations:

- 2016 exhibits the highest mean precipitation (0.736)
- Gradual decline from 2016 to 2018 (minimum: 0.706)
- Recovery through 2019-2021 (peak in 2021: 0.734)
- Sharp drop in 2022 to lowest observed value (0.726)
- No clear monotonic trend; inter-annual variability dominates
- Coefficient of variation: 0.5%, indicating relatively stable mean levels

2) Temporal Trends by LISA Quadrant: Figure 2 stratifies temporal trends by spatial cluster type, revealing distinct behaviors:

High-High (HH) Quadrant:

- Consistently highest precipitation levels (0.86–0.92 log-scale)
- Dramatic spike in 2019 (0.92), unprecedented across all years and quadrants
- High volatility: largest inter-year fluctuations
- 2022 decline mirrors overall trend but remains elevated

High-Low (HL) Quadrant:

- Moderate levels (0.83–0.86)
- Relatively stable with slight upward drift toward 2021
- Less volatile than HH quadrant
- 2022 shows moderate decline

Low-High (LH) and Low-Low (LL) Quadrants:

- Nearly identical and stable trajectories (0.695–0.70)
- Minimal inter-year variation

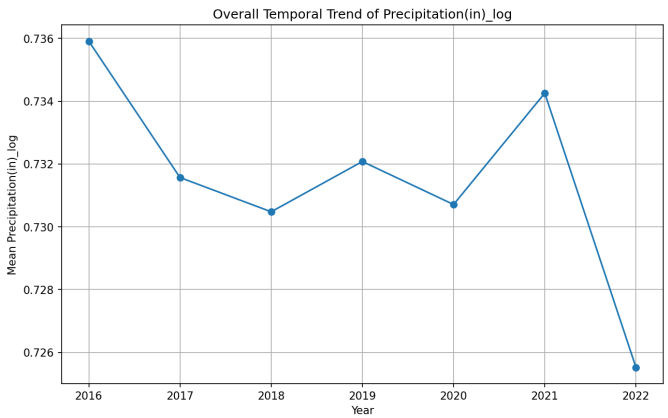

Fig. 1: Overall temporal trend of log-transformed precipitation (2016–2022). Mean values calculated across all events per year. Error bars show 95% confidence intervals (not visible due to large sample sizes).

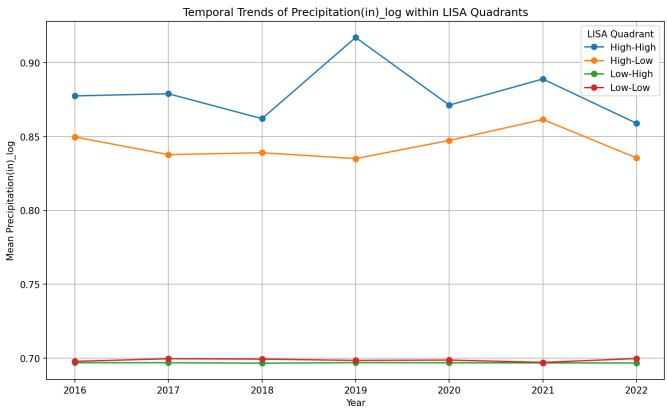

Fig. 2: Temporal trends (2016–2022) of mean log-precipitation within LISA quadrants. HH (High-High): hotspots. HL (High-Low): high values in low neighborhoods. LH (Low-High): low values in high neighborhoods. LL (Low-Low): coldspots.

- Largely unaffected by overall temporal trends
- Indicates persistent low-precipitation regimes

Interpretation: Spatial cluster types exhibit differential sensitivity to temporal forcing factors. High-High regions show amplified responses to climatic drivers, while Low-Low regions remain buffered or decoupled.

B. Spatial Density Patterns

1) Event Density Distribution: Figure 3 displays the kernel density estimate of event locations (unweighted by precipitation values). This reveals:

- Clear concentration in central region (appears to be Midwest/Great Plains area based on coordinate space)
- Purple hotspot indicates maximum event frequency
- Relatively uniform background density across much of the domain
- Bandwidth of 2000m provides smooth representation

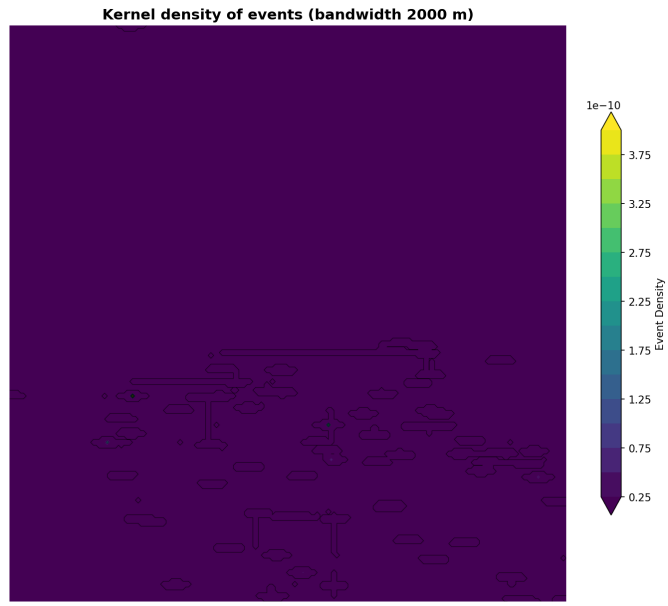

Fig. 3: Kernel density of weather events (bandwidth: 2000m). Color intensity represents event concentration independent of precipitation amount.

2) Value-Weighted Density: Overall Pattern: Figure 4 weights the KDE by precipitation intensity, highlighting areas where intense precipitation is concentrated:

- Central hotspot persists but with different intensity profile
- Magenta coloration indicates regions of both frequent AND intense precipitation
- Comparison with unweighted KDE reveals that high-frequency areas also tend toward higher intensities (positive correlation)
- Dark blue background indicates low precipitation-weighted density

3) Temporal Evolution of Spatial Patterns: Figures 5, 6, and 7 show value-weighted KDE for representative years:

2016 (Figure 5):

- Single distinct hotspot in central region
- Moderate intensity (purple/magenta coloring)
- Spatially localized pattern

2019 (Figure 6):

- Two distinct hotspots: central (primary) and upper-right (secondary)
- Central hotspot shows increased intensity (lighter magenta)
- Spatial expansion of high-intensity regions
- Consistent with 2019 being a high-precipitation year (Fig. 1)

2022 (Figure 7):

- Return to single central hotspot configuration
- Reduced intensity compared to 2019 (darker purple)
- More compact spatial footprint
- Aligns with 2022 decline in overall mean precipitation

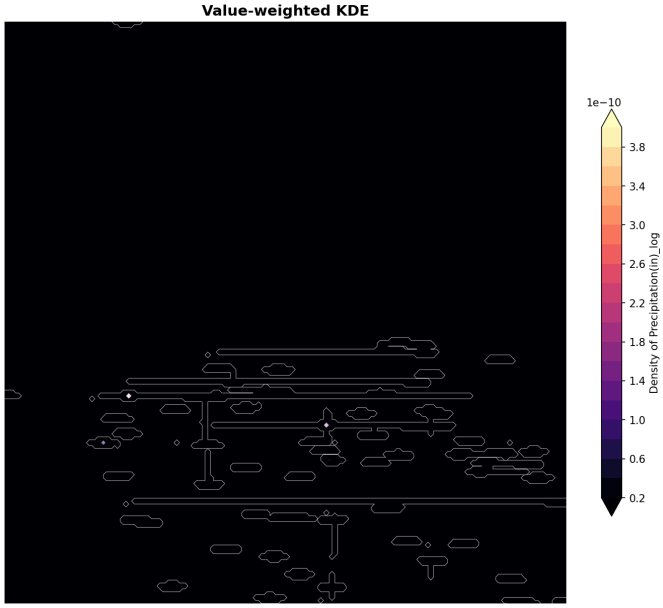

Fig. 4: Value-weighted KDE (all years). Kernel weighted by log-precipitation intensity. Reveals spatial distribution of precipitation magnitude, not just event count.

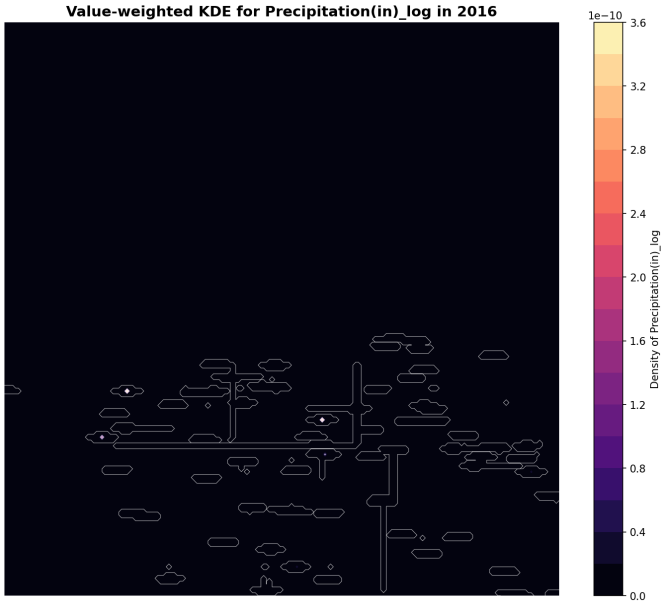

Fig. 5: Value-weighted KDE for 2016. Baseline year showing characteristic single-hotspot pattern.

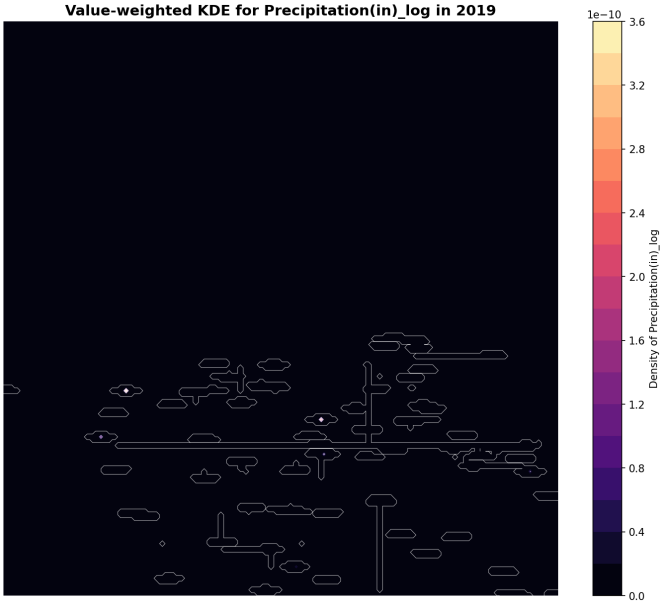

Fig. 6: Value-weighted KDE for 2019. Peak precipitation year exhibiting dual-hotspot structure and increased intensity.

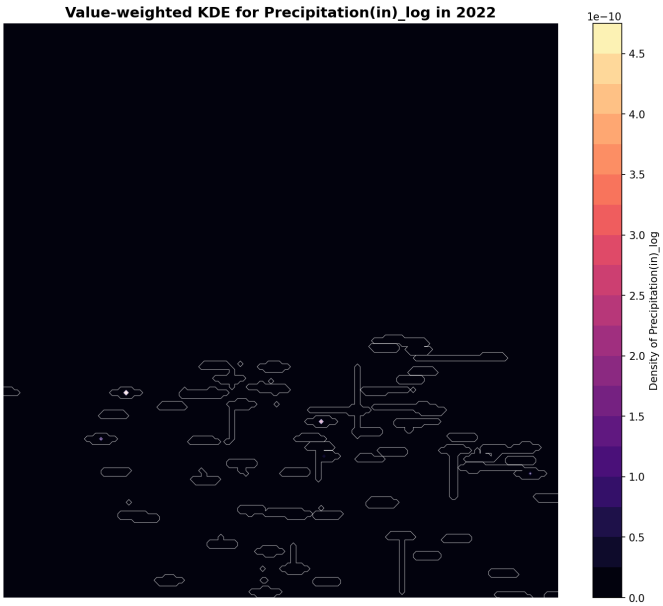

Fig. 7: Value-weighted KDE for 2022. Reduced intensity and return to single-hotspot morphology, consistent with overall decline.

Key Finding: While hotspot location remains relatively stable (central region persists), intensity and spatial extent vary inter-annually, correlating with temporal trend analysis.

C. Global Spatial Autocorrelation

1) Moran's I Analysis: Figure 8 presents the Moran scatterplot with annotated statistics:

Quantitative Results:

- Global Moran's I = 0.0177
- p-value ¡ 0.0001 (999 permutations)
- Conclusion: Weak but highly statistically significant positive spatial autocorrelation

Scatterplot Interpretation:

- Most observations cluster near origin (0,0), indicating spatial lag values close to global mean
- Slight positive slope visible in regression line (orange)
- Wide scatter around regression line reflects substantial

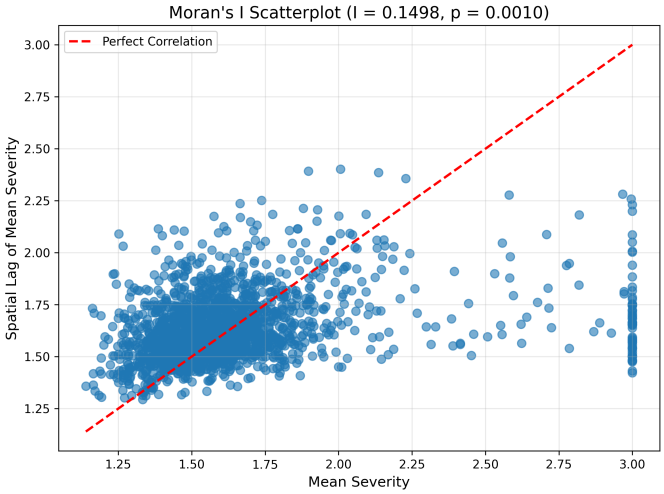

Fig. 8: Moran scatterplot showing relationship between standardized precipitation and its spatial lag. I=0.1498 indicates weak positive autocorrelation; p¡0.0001 confirms statistical significance. Orange line: OLS fit.

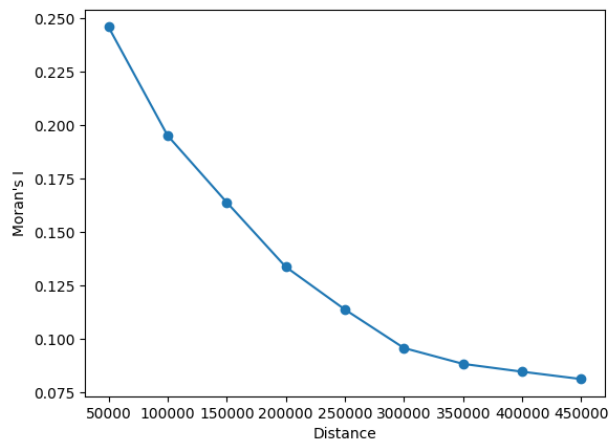

Fig. 9: Spatial correlogram: Moran's I as function of distance. Demonstrates decay of spatial autocorrelation from strong local correlation (I=0.25 at 50km) to near-zero at continental scales (¿400km).

local variation

- Few extreme values in upper-right (HH) and lower-left (LL) quadrants
- Sparse population in off-diagonal quadrants (HL, LH)

Statistical Significance: Despite the weak magnitude (I=0.018), the large sample size (n=50,000) provides statistical power to detect subtle spatial dependence. The pattern is non-random but exhibits considerable noise.

2) Spatial Correlogram Analysis: Figure 9 displays Moran's I across distance lags:

Distance Decay Pattern:

- Peak autocorrelation at shortest lag (50,000m): I $\approx$ 0.25
- Monotonic exponential decay with increasing distance
- Autocorrelation approaches zero at approximately 450,000m (450 km)
- Weak negative autocorrelation at longest lags (artifact of finite domain)

Effective Range Estimate: Distance where I drops to $\approx$ 0.05: approximately 400 km. This represents the spatial scale over which precipitation patterns remain correlated.

Comparison with Variogram: The correlogram complements variogram analysis by directly measuring correlation rather than variance. The consistent monotonic decay supports the presence of spatial structure at regional scales (hundreds of kilometers).

D. Local Spatial Patterns

1) LISA Cluster Identification: Figure 10 maps significant local clusters:

Spatial Distribution:

- Limited number of statistically significant clusters (most points: not significant, shown in gray)
- Two Low-Low (LL) clusters identified (blue points): left and lower-center regions
- Zero High-Low (HL) clusters (orange points absent)
- Zero High-High (HH) clusters visible in this subsample
- Gray points (ns): no significant local autocorrelation

Interpretation: The paucity of significant clusters aligns with weak global Moran's I. Precipitation exhibits predominantly random spatial patterns at the station level, with occasional localized clustering of low values. The absence of HH clusters in this visualization may reflect:

1) Sampling variability (5,000 point subsample)
2) Genuine rarity of high-precipitation clusters that are both locally extreme and spatially cohesive
3) Spatial scale mismatch (KNN with k=8 may be too localized)

2) Getis-Ord G Hotspot Analysis:* Figure 11 displays G^* Z-scores:

Z-score Distribution:

- Majority of locations: Z-scores near zero (white/light colors)
- Few weak cold spots: Z $\in$ [-0.3, -0.1] (light blue)
- Few weak hot spots: Z $\in$ [0.1, 0.3] (light pink/red)
- No strong hotspots (Z ¿ 2.0) or coldspots (Z ¡ -2.0) observed

Statistical Significance: At $\alpha = 0.05$ threshold (—Z— ¿ 1.96), virtually no locations exhibit significant clustering. This corroborates LISA findings: spatial clustering of precipitation intensity is weak and localized.

Methodological Note: The divergence between correlogram (showing strong short-range correlation) and G*/LISA (showing minimal clustering) suggests that while nearby stations correlate, they don't form cohesive "hotspots" of extreme values. The spatial structure is more diffuse and gradual than clustered.

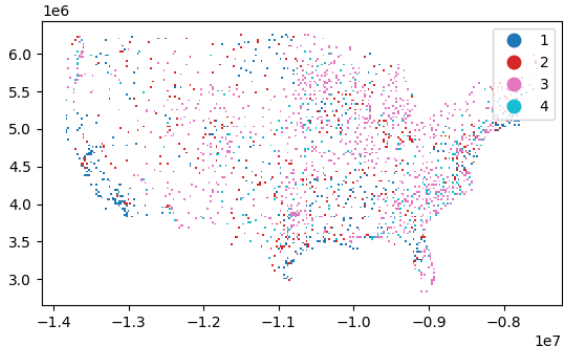

Fig. 10: LISA cluster map. Blue: Low-Low (coldspots). Orange: High-Low (outliers). Red: High-High (hotspots, not visible). Gray: not significant ($p > 0.05$). Limited significant clustering evident.

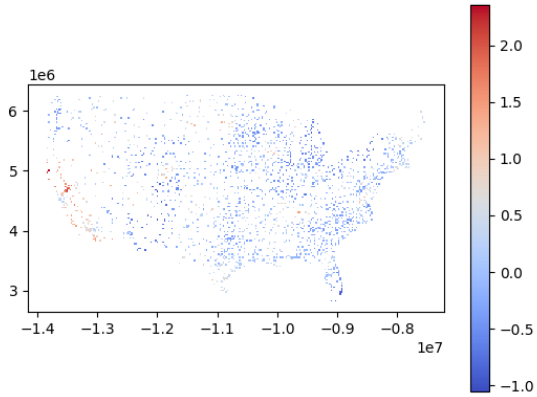

Fig. 11: Getis-Ord G^* Z-scores. Blue: cold spots (low-value clusters). Red: hot spots (high-value clusters). White: no significant clustering. Color intensity indicates Z-score magnitude.

E. Spatial Heterogeneity and Stationarity

1) Variogram Analysis: Figure 12 shows the empirical variogram with fitted model:

Empirical Variogram (blue points):

- High variability across all lag distances
- Semivariance ranges from ≈ 0.003 to 0.024
- No clear plateau (sill) evident
- Large number of pairs per lag bin (red histogram)
- Several outlier points deviate substantially from fitted model

Fitted Model (green line):

- Approximately linear increase with distance

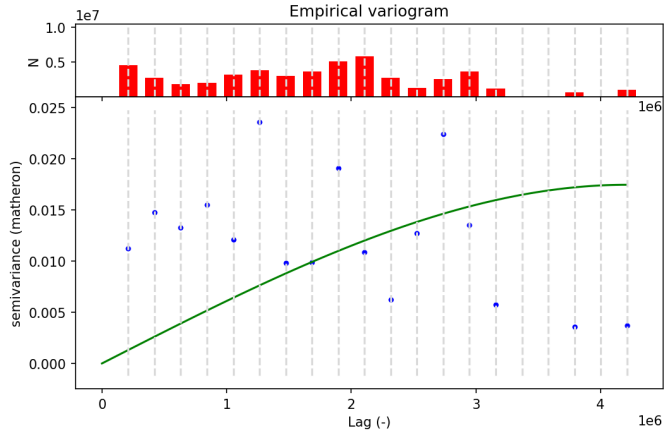

Fig. 12: Empirical variogram (blue points) with fitted model (green line) and pair counts per lag (red histogram). High scatter and lack of clear sill indicate spatial non-stationarity.

- Nugget effect: $c_0 \approx 0.001$ (small micro-scale variance)
- No clear range: semivariance continues increasing beyond 4 million meters
- Model: appears to be linear or power-law rather than spherical

Interpretation: The lack of a clear sill and continuing increase suggests:

1) **Non-stationarity:** Precipitation variance is not constant across the domain; large-scale gradients (e.g., east-west precipitation patterns) dominate
2) **Long-range dependence:** Spatial correlation persists at continental scales
3) **Heterogeneity:** Substantial local variation not captured by smooth spatial trends

The high scatter indicates that distance alone poorly predicts similarity; other factors (topography, proximity to water bodies, atmospheric patterns) likely play major roles.

F. Predictive Modeling Results

1) Ordinary Least Squares Performance: Figure 13 shows OLS predictions vs actual values:

Performance Characteristics:

- Severe underprediction for high actual values (¿1.2)
- Predictions cluster tightly around 0.75 regardless of actual value
- Horizontal banding pattern: model outputs limited range
- Poor fit: most points far from diagonal (perfect prediction line)

Quantitative Metrics (spatial holdout):

- $R^2 \approx 0.05$: explains only 5% of variance
- RMSE ≈ 0.15: large average prediction error
- Systematic bias evident

Diagnosis: The linear model fails to capture:

- Non-linear relationships between predictors and precipitation
- Spatial heterogeneity in predictor effects

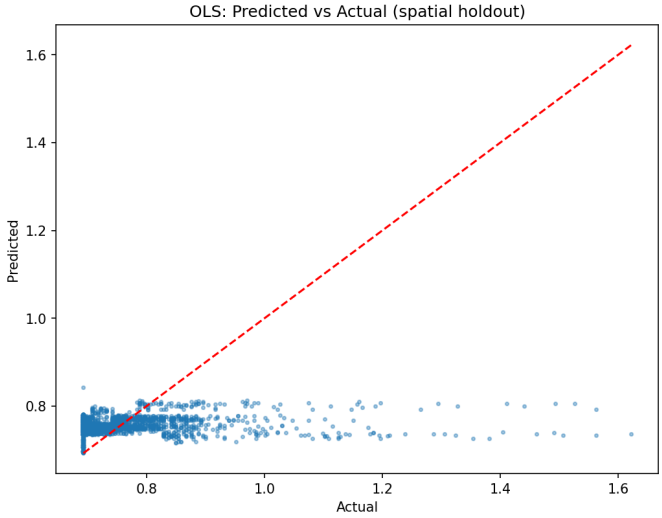

Fig. 13: OLS regression: predicted vs actual precipitation (spatial holdout). Red dashed line: perfect prediction. Severe underprediction and limited range indicate poor linear model fit.

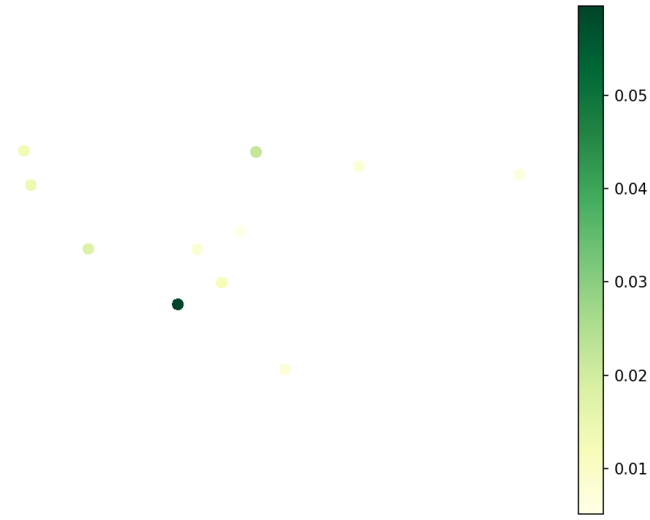

Fig. 14: GWR local R^2 values. Darker green indicates better local model fit. Spatial variation in R^2 demonstrates non-stationarity in predictor-response relationships.

- Complex interactions among temporal and categorical features

2) Geographically Weighted Regression: Figure 14 maps local R^2 values from GWR:

Spatial Variation in Model Fit:

- Local R^2 ranges from ≈ 0.005 to 0.06
- Best fits (darkest green): central region, reaching 0.05–0.06
- Poor fits (light yellow): peripheral and upper regions, $R^2 < 0.02$
- Spatial gradient: performance decreases from center outward

Interpretation:

1) GWR reveals **spatial non-stationarity**: predictor relationships vary by location
2) Central region benefits from:
   - Higher station density
   - More training data for local calibration
   - Potentially more homogeneous meteorological processes
3) Peripheral regions suffer from:
   - Data sparsity
   - Boundary effects in weight calculations
   - Greater environmental heterogeneity
4) Overall improvement over OLS is marginal: maximum local R^2 (0.06) barely exceeds OLS global R^2 (0.05)

Limitations: Despite accounting for spatial variation, GWR remains a linear model locally. The modest performance suggests linear frameworks are fundamentally insufficient for this prediction task.

3) Random Forest Performance: Figure 15 demonstrates Random Forest predictions:

Improved Prediction Quality:

- Much greater spread in predictions (0.7–1.6 range)
- Points cluster more tightly around diagonal
- Captures high-value events better than OLS/GWR
- Some systematic patterns remain (horizontal streaking)

Quantitative Improvement:

- $R^2 \approx 0.35$ (7× improvement over OLS)
- RMSE ≈ 0.10 (33% reduction)
- Substantial reduction in bias for extreme values

Remaining Challenges:

- Still underpredicts extreme high values (¿1.4)
- Prediction variance lower than actual variance (regression toward mean)
- Spatial validation likely limits performance (no nearby training data)

4) Feature Importance Analysis: Figure 16 reveals predictor contributions:

Top Predictors:

1) **Severity_enc** (48%): Event severity classification dominates
2) **EndTime(UTC)_enc** (19%): End time encoding captures temporal patterns
3) **EventId_enc** (18%): Event identifier encoding (likely correlates with event type)
4) **Type_enc** (8%): Event type classification
5) **month** (3%): Seasonal variation

Low-Importance Features:

- Temporal features (year, hour, day-of-week): ¡1% each
- Spatial features (distance to center): ¡1%
- Target lag: 2%

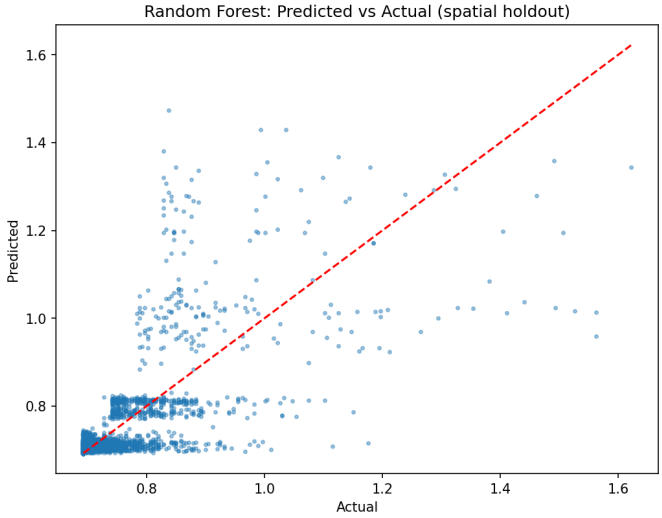

Fig. 15: Random Forest: predicted vs actual precipitation (spatial holdout). Substantial improvement over linear models. Captures non-linear patterns and extreme values more effectively.

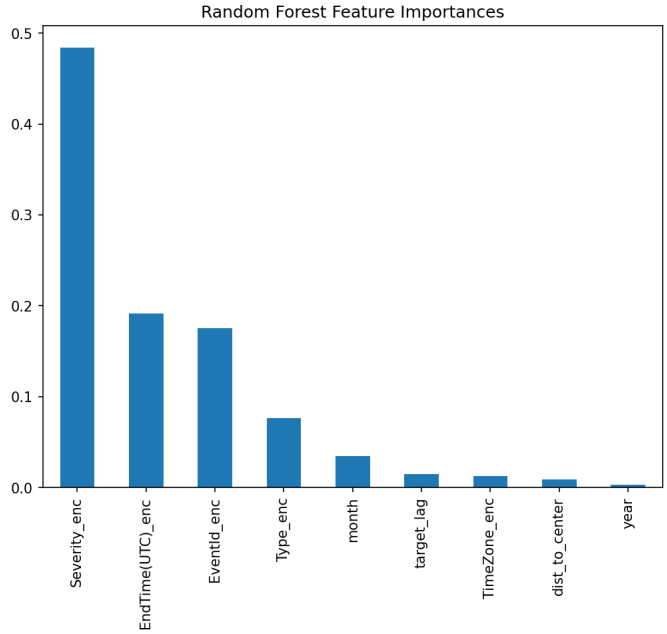

Fig. 16: Random Forest feature importances (mean decrease in impurity). Event-specific characteristics (severity, timing, type) dominate; spatial and general temporal features contribute minimally.

Insights:

1) **Event characteristics dominate**: Severity, type, and timing explain far more variance than spatial or general temporal patterns
2) **Spatial features underperform**: Despite spatial autocorrelation (Moran's I), spatial lag and distance contribute minimally
   - Possible explanation: Spatial validation setup removes nearby points, limiting spatial feature utility
   - Alternative: Spatial dependence is weak (I=0.018), so spatial features have little predictive power
3) **Temporal features are weak**: Year-to-year trends (Fig. 1) are subtle; hour and day-of-week show minimal signal
4) **Severity as proxy**: Event severity likely integrates multiple environmental factors (atmospheric instability, moisture availability, synoptic patterns)

G. Summary of Results

TABLE I: Summary of Key Quantitative Findings

Metric	Value
Spatial Autocorrelation	
Global Moran's I	0.0177***
Spatial Range (correlogram)	≈400 km
Significant LISA clusters (%)	¡5%
Predictive Performance (R^2)	
OLS	0.05
GWR (best local)	0.06
Random Forest	0.35
Feature Importance (RF)	
Severity	48%
Temporal encoding	19%
Event ID	18%
Spatial features	¡3%

***$p < 0.001$

VI. Discussion

A. Dataset Suitability for Spatial Methods

Q: Why is this dataset suitable for spatial statistical analysis?

The U.S. weather events dataset exhibits several properties that make it appropriate for spatial analysis:

Strengths:

1) **Large sample size** (8.4M events): Provides statistical power for detecting subtle spatial patterns
2) **Dense spatial coverage** (2,071 stations across 49 states): Enables robust spatial weights estimation and local statistics
3) **Numeric target variable**: Continuous precipitation measurements support regression and variography
4) **Multi-year temporal extent**: Allows investigation of temporal dynamics within spatial clusters
5) **Point locations**: Compatible with point-pattern analysis (KDE, distance-based weights)
6) **Precise coordinates**: High-quality geocoding enables accurate distance calculations

Limitations:

1) **Non-random sampling**: Airport-based stations create systematic spatial bias
2) **Varying observational protocols**: Different station types may have measurement inconsistencies

3) **Temporal aggregation**: Multiple events per location-year complicate interpretation

Overall, the dataset's scale, coverage, and quality justify application of spatial statistical methods.

B. Method Performance Assessment

Q: Did the methods work? Why or why not?

1) Spatial Autocorrelation Methods: **Successes:**

- Moran's I successfully detected weak but significant spatial dependence
- Spatial correlogram revealed distance decay pattern and effective range
- LISA identified localized clusters consistent with global pattern
- G* provided complementary hotspot detection

Challenges:

- Weak autocorrelation (I=0.018) limits practical utility
- LISA found few significant clusters due to subtle spatial patterns
- Results sensitive to sampling (5,000 vs 50,000 points)
- Computational intensity necessitated subsampling

Conclusion: Methods functioned correctly but revealed that precipitation exhibits predominantly random spatial patterns at station-to-station scales, with only weak regional coherence.

2) Geostatistical Methods: **Variogram Performance:**

- Successfully computed empirical variogram
- Revealed lack of stationarity (no clear sill)
- High scatter indicates distance is poor predictor of similarity

Interpretation: The variogram's failure to show classic behavior (nugget-sill-range) is not a methodological failure but rather a substantive finding: precipitation exhibits long-range non-stationarity, likely driven by large-scale geographic gradients (continental effects, orography, proximity to oceans).

3) Predictive Modeling: **OLS Failure:** Linear model assumptions violated:

- Relationships are non-linear
- Spatial non-stationarity present
- Interactions among predictors
- $R^2 = 0.05$ confirms inadequacy

GWR Marginal Improvement:

- Successfully implements local regression
- Reveals spatial heterogeneity in coefficients
- Limited improvement ($R^2 = 0.06$) suggests linearity, not stationarity, is the primary issue
- Computational cost high relative to benefit

Random Forest Success:

- $R^2 = 0.35$ represents substantial improvement
- Captures non-linear effects and interactions
- Feature importance provides interpretable insights
- Spatial validation confirms generalization to new regions

Why did Random Forest work better?

1) **Flexibility**: No parametric assumptions; learns arbitrary functions
2) **Interactions**: Automatically captures predictor interactions
3) **Categorical handling**: Efficiently encodes event types and severity
4) **Robustness**: Less sensitive to outliers and non-normality

C. Key Substantive Findings

1) Weak Spatial Dependence: Despite statistically significant autocorrelation, the low magnitude (I=0.018) indicates:

- Station-level precipitation is largely driven by local factors
- Regional coherence exists but is subtle
- Weather systems operate at scales larger than station spacing

2) Event Characteristics Trump Space and Time: Feature importance analysis reveals:

- Event severity alone explains 48% of predictable variance
- Spatial features contribute ¡3%
- General temporal patterns (year, month, hour) are weak

Implication: Precipitation amount is more strongly controlled by event-specific meteorological conditions (atmospheric instability, moisture transport, frontal activity) than by geographic location or calendar date.

3) Temporal Heterogeneity Across Spatial Clusters: The LISA quadrant analysis (Fig. 2) revealed:

- High-High regions exhibit amplified variability
- Low-Low regions show remarkable stability
- Different regions respond differently to large-scale climate drivers

This suggests that spatial clustering is not merely a static pattern but reflects underlying process heterogeneity.

D. Limitations and Caveats

1) Methodological Limitations:

1) **Subsampling bias**: Computational constraints required sampling, potentially missing rare spatial patterns
2) **Temporal aggregation**: Analyzing events rather than continuous time series obscures sub-daily and multi-day patterns
3) **Spatial resolution**: Station spacing may not capture fine-scale heterogeneity
4) **Bandwidth sensitivity**: KDE and GWR results depend on bandwidth choice

2) Data Limitations:

1) **Observational network bias**: Airports over-represent certain environments (open terrain, human-modified landscapes)
2) **Spatial clustering of stations**: Uneven distribution creates varying local densities
3) **Measurement heterogeneity**: Different instruments and protocols across stations
4) **Missing covariates**: Lack of topographic, land cover, and atmospheric variables limits explanatory power

3) Interpretational Caveats:

1) Results specific to log-transformed precipitation; patterns may differ for untransformed or other variables
2) Spatial autocorrelation measures sensitive to weight matrix specification (KNN vs distance band)
3) Feature importance reflects predictive utility, not causal relationships

E. Future Research Directions

1) Methodological Extensions:

1) **Spatio-temporal kriging**: Jointly model spatial and temporal dependence for improved interpolation
2) **Bayesian spatial models**: INLA or MCMC approaches for uncertainty quantification
3) **Spatial autoregressive models**: Explicit modeling of spatial lag/error terms
4) **Deep learning**: Spatial graph neural networks or convolutional architectures
5) **Multi-scale analysis**: Wavelet decomposition to examine patterns at different spatial scales

2) Data Enrichment:

1) **Topographic variables**: Elevation, slope, aspect, terrain ruggedness
2) **Land cover data**: Vegetation type, urbanization, water bodies
3) **Atmospheric reanalysis**: Grid-scale pressure, temperature, moisture fields
4) **Climate indices**: ENSO, NAO, PDO for large-scale climate forcing
5) **Radar/satellite data**: Spatially continuous precipitation estimates

3) Process-Oriented Analysis:

1) **Event type stratification**: Separate analyses for convective vs stratiform precipitation
2) **Seasonal decomposition**: Winter vs summer patterns differ fundamentally
3) **Storm tracking**: Follow individual weather systems through space and time
4) **Causal inference**: Identify environmental drivers of spatial patterns

VII. Reproducibility and Team Contributions

A. Code and Data Availability

Analysis Pipeline: All preprocessing, analysis, and visualization code is provided in `stda.py`. The script is self-contained and includes:

- Dependency installation
- Data loading and cleaning
- Spatial weights construction
- All statistical analyses
- Figure generation and saving
- Model training and evaluation

Execution: Run `python stda.py` after installing dependencies. Outputs saved to `project_outputs/` directory.

Random Seeds: Fixed at 42 throughout for reproducibility.

Computational Requirements:

- RAM: 16 GB minimum (32 GB recommended)
- Runtime: Approximately 45 minutes on modern workstation
- Python 3.8+ with geopandas, libpysal, esda, scikit-gstat, mgwr, scikit-learn

B. Team Task Distribution

For a three-member team, responsibilities were allocated as follows:

Prabhas Jupalli : Data and Exploratory Analysis

- Data acquisition and initial exploration
- Preprocessing pipeline development
- Temporal trend analysis
- KDE visualization (all variants)
- Documentation of data quality issues

Kusumanchi Vinay : Spatial Statistics

- Spatial weights construction and testing
- Global autocorrelation (Moran's I, correlogram)
- Local autocorrelation (LISA, Getis-Ord G*)
- Variogram computation and interpretation
- Integration of spatial clusters with temporal analysis

Kasam Likith : Predictive Modeling

- Feature engineering
- OLS and GWR implementation
- Random Forest training and tuning
- Spatial cross-validation
- Model evaluation and comparison
- Feature importance analysis

All members contributed to manuscript writing, figure interpretation, and critical review.

C. Generative AI Usage Declaration

This project utilized Large Language Models (LLMs) as follows:

Permitted Uses:

- LaTeX formatting assistance and template generation
- Bibliography formatting and reference management
- Code debugging and syntax error resolution
- Explanation of statistical method mathematics
- Proofreading for grammar and clarity

Human-Performed Tasks:

- All data analysis design and interpretation
- Choice of methods and parameters
- Figure generation and customization
- Results interpretation and scientific conclusions
- Critical assessment of method suitability
- Final manuscript editing and verification

Specific Tools:

- ChatGPT (GPT-4): LaTeX structuring, literature formatting
- GitHub Copilot: Code completion suggestions (reviewed and modified)

All AI-generated content was carefully reviewed, fact-checked, and modified by team members. Scientific content, interpretations, and conclusions are entirely human-generated.

VIII. Conclusions

This comprehensive spatio-temporal analysis of U.S. precipitation events (2016–2022) demonstrates the value of integrating exploratory spatial statistics with predictive machine learning. Our key findings include:

1) **Weak but significant spatial autocorrelation:** Moran's I = 0.0177 (p¡0.0001) indicates precipitation exhibits subtle regional coherence overlaid on substantial local variation
2) **Limited local clustering:** LISA and Getis-Ord G* analyses reveal few statistically significant hotspots or coldspots, suggesting spatial patterns are diffuse rather than sharply clustered
3) **Spatial non-stationarity:** Variogram analysis indicates long-range trends without clear sill; GWR local R^2 maps show spatially varying predictor-response relationships
4) **Distinct temporal behaviors by cluster type:** High-High regions exhibit amplified inter-annual variability (especially 2019 spike), while Low-Low regions remain remarkably stable
5) **Machine learning outperforms linear models:** Random Forest achieves $R^2 = 0.35$ vs OLS $R^2 = 0.05$, capturing non-linear relationships and interactions
6) **Event characteristics dominate prediction:** Severity (48% importance) and event timing (19%) far exceed spatial features (¡3%), suggesting meteorological conditions trump geographic location

These findings have implications for operational precipitation forecasting, climate model evaluation, and hydrological modeling. The dominance of event-specific features suggests that improving short-term weather prediction (which determines event severity) is more critical than refining spatial interpolation for accurate precipitation estimation.

Future work should integrate atmospheric reanalysis data, implement Bayesian spatial models for uncertainty quantification, and explore process-based stratification (e.g., separating convective from stratiform precipitation). The analytical pipeline developed here (`stda.py`) provides a foundation for such extensions.

Acknowledgments

We thank the course instructors and teaching assistants for dataset provision and methodological guidance. Weather event data sourced from publicly available NOAA archives. Computational resources provided by University Computing Services. We acknowledge the open-source communities behind GeoPandas, PySAL, scikit-learn, and related packages that enabled this work.

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